User Stories for Record/Replay

ID	User Story Name	User Story Description	Priority	Estimated Effort (hours)
11	Save moves	As a player, I want the game to record each move I've made.	1	3
12	Replay moves	As a player, I want to be able to replay the game up to my current state.	2	6

Acceptance Criteria for Record/Replay

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status
11 - Save Moves	11.1	**AC 11.1 <Save manual moves>** Given the game is not in replay mode When I make a valid manual move Then the move should be saved to the record	Complete
	11.2	**AC 11.2 <Save automated moves>** Given the game is not in replay mode When I press the automated move button And a move is successfully made Then the random move should be saved to the record	Complete
	11.3	**AC 11.3 <Do not record moves in replay mode>** Given the game is in replay mode When a manual or automated move is made Then the move should NOT be added to the record And the record should remain exactly the same	Complete
	11.4	**AC 11.4 <Do not record invalid manual moves>** Given the game is not in replay mode When I attempt to make a manual move And the move is not valid Then the move should NOT be added to the record And the record should remain exactly the same	Complete
	11.5	**AC 11.5 <Do not record automated moves when game is over>**	Complete

		Given the game is not in replay mode And the game is over When I click the automated move button Then the record should remain exactly the same	
	11.6	**AC 11.6 <Wipe the record>** At any time When I press the reset button, change board size, change board type, or randomize Then the record should be wiped clean	Compl ete
	11.7	**AC 11.7 <Record when randomized>** At any time When I press the randomize button Then the record should be wiped clean And the randomized board should be saved to the record	Compl ete
12 - Replay Moves	12.1	**AC 12.1 <Replay game>** Given there are moves on the record And I have not randomized since the previous board reset When I press the replay button Then the board should reset And all moves up until now should play on their own, one by one And selected squares should be unselected	Compl ete
	12.2	**AC 12.2 <Replay from randomized board>** Given there are moves on the record And the board has not been reset or reconfigured since its last randomization When I press the replay button Then the board should revert to its randomized configuration And all the moves until now since then should play on their own, one by one.	Compl ete
	12.3	**AC 12.3 <Interrupt replay>** Given the game is in replay mode When I change board size, change board type, reset, or randomize Then replay mode should end immediately And no more recorded moves should play And the record should be wiped	Compl ete
	12.4	**AC 12.4 <Loop replay if already replaying>**	Compl

		Given the game is in replay mode When I press the replay button Then the board should reset And all moves up until now should play on their own, one by one And the previous replay operation should terminate	ete
	12.5	**AC 12.5 <Don't replay if no moves have been made>** Given that no moves have been recorded When I press the replay button Then nothing should happen And the player should get an alert	Compl ete
	12.6	**AC 12.6 <Resume game when replay ends>** Given that there are moves on the record And we are in replay mode When all moves on the record are made in replay mode And the moves have not ended the game Then the game should return to play mode And new moves should continue to be added to the record	Compl ete
	12.7	**AC 12.7 <No alerts if replay ends game>** Given that we are in replay mode When all moves on the record are made in replay mode And there are no more valid moves Then the game should NOT show the usual game over alert And the game should return to play mode	Compl ete
	12.8	**AC 12.8 <Don't allow cell selection in replay mode>** Given the board is in replay mode When the player attempts to click a cell Then no cell should be selected And the board state should remain the same And replay should continue uninterrupted	Compl ete
	12.9	**AC 12.9 <Don't allow autoplay in replay mode>** Given the board is in replay mode When the player clicks the autoplay button Then nothing should happen And replay should continue uninterrupted	Compl ete

Updated Acceptance Criteria

User Story ID and Name	AC ID	Description of Acceptance Criteria	Status (completed, toDo, inProgress)
1.Choose board type and size	1.1	**AC 1.1 <Select board type>** Given I am on the main menu When I select option English, Hexagon, or Diamond Then the selected board type should be stored	complete
	1.2	**AC 1.2 <Select board size>** Given I am on the main menu When I input a board size And board size is within valid range for its type Then the inputted size should be stored	complete
	1.3	**AC 1.3 <Reject invalid board sizes>** Given I am on the main menu And I've inputted a board size out of acceptable range When I try to start a new game Then I receive an error message And no board is generated	complete
	1.4	**AC 1.4 <Default configuration when opening program>** When I boot the program Then the board size and type are set to default configurations	complete
2. Beginning a game	2.1	**AC 2.1 <Beginning a game with default board configuration>** Given I am on the main menu And I am on default board configuration When I start a new game Then the game board should be displayed with default configuration And all board positions should contain pegs except the center position And the center position should be empty And the game state should be "in progress"	complete
	2.2	**AC 2.2 <Beginning a game applies the selected board configuration>**	complete

		Given I am on the main menu And I have selected a specific board type and size When I start a new game Then the board should be displayed And the board should match the selected type And the board should match the selected size And all board positions should contain pegs except the center position And the center position should be empty And the game state should be “in progress”	
3. View the Current Board State	3.1	**AC 3.1 <Display the current state of the board after a successful move>** Given the game is in progress When I make a valid move Then the GUI should update the board to show the new game state	com plete
4. Make a Move	4.1	**AC 4.1 <Make a Move>** Given the game is in progress And a peg is located at a valid board position And an adjacent position contains another peg And the position two spaces away in the same direction is empty When the player selects the peg and performs a jump to the empty position Then the peg moves to the empty position And the jumped peg is removed from the board And the original position becomes empty And the board state is updated	com plete
	4.2	**AC 4.2 <Reject Selection of Empty Position>** Given the game is in progress When the player attempts to select a position with no peg Then the selection does not occur	com plete
	4.3	**AC 4.3 <Reject moves that jump over an empty position>** Given the game is in progress And a peg is selected And the adjacent position in the selected direction is empty	com plete

		When the player attempts to make a move Then the move is rejected And the game state remains unchanged	
	4.4	**AC 4.4 <Reject moves that land in an occupied position>** Given the game is in progress And a peg is selected And the landing position is occupied When the player attempts to make a move Then the move is rejected And the game state remains unchanged	com plete
	4.5	**AC 4.5 <Reject moves that jump more than one peg>** Given the game is in progress And a peg is selected And there is more than one peg between the start and landing positions When the player attempts to make a move Then the move is rejected And the game state remains unchanged	com plete
	4.6	**AC 4.6 <Reject moves that put a peg outside the board boundaries>** Given the game is in progress And the landing position is outside the board When the player attempts to make a move Then the move is rejected And the game state remains unchanged	com plete
	4.7	**AC 4.7 <Reject moves in non-allowed directions>** Given the game is in progress And the selected move is not orthogonal or diagonal When the player attempts to make a move Then the move is rejected And the game state remains unchanged	com plete
	4.8	**AC 4.8 <Only process one move at a time>** Given the game is in progress When the player completes a valid move Then no additional moves are processed until the current move is finished	com plete

	4.9	**AC 4.9 <Reject moves to neighboring positions>** Given the game is in progress And a peg is selected And the landing position is neighbors to the selected position When the player attempts to make a move Then the move is rejected And the game state remains unchanged	com plete
5. Detect Game Over	5.1	**AC 5.1 <Detect when no more valid moves>** Given the game is in progress When the player has made a valid move And there are no longer any valid moves available Then the game should end And the player can no longer make moves	com plete
	5.2	**AC 5.2 <Detect bugged start>** When a new game begins And, somehow, there are no valid moves to make Then the game should end And the player can no longer make moves	canc elled
6. Determine Performance	6.1	**AC 6.1 <Rank based on remaining pegs>** Given the game is in progress When the game ends Then the player should receive a ranking based on the number of pegs remaining	Canc elled
7. Reset	7.1	**AC 7.1 <Reset during game>** Given the game is in progress When the player chooses to reset Then the board should return to its original state (all pegs filled except the center) And the game state should reset too	com plete
	7.2	**AC 7.2 <Reset after game>** Given the game has ended When the player chooses to reset Then the board should return to its original state (all pegs filled except the center) And the game state should reset too	com plete

8. Reconfigure Board	8.1	**AC 8.1 <Reconfigure during the game>** Given the game is in progress When the player chooses to start a new game Then we delete the current game state and board And return to the main menu, where the player can choose board size and type	com plete
	8.2	**AC 8.2 <Reconfigure after the game>** Given the game is over When the player chooses to start a new game Then we delete the current game state and board And return to the main menu, where the player can choose board size and type	com plete
9. Automated moves	9.1	**AC 9.1 <Make a random move>** Given the game is ongoing When the player presses the autoplay button Then the board makes a random valid move	com plete
	9.2	**AC 9.2 <Don't make automated move if game over>** Given the game is over When the player presses the autoplay button Then there should be no change to game state	com plete
10. Randomize board	10.1	**AC 10.1 <Randomize board>** When the player presses the randomize button Then every valid space should randomly become either an empty square or a square with a peg	com plete
	10.2	**AC 10.2 <Don't randomize invalid cells>** When the player presses the randomize button Then all cells which are invalid should remain invalid	com plete
	10.3	**AC 10.3 <Randomized board must be valid>** When the player presses the randomize button Then the randomized board should be one where valid move(s) are available	com plete
11 - Save Moves	11.1	**AC 11.1 <Save manual moves>** Given the game is not in replay mode When I make a valid manual move Then the move should be saved to the record	Co mpl ete

	11.2	**AC 11.2 <Save automated moves>** Given the game is not in replay mode When I press the automated move button And a move is successfully made Then the random move should be saved to the record	Complete
	11.3	**AC 11.3 <Do not record moves in replay mode>** Given the game is in replay mode When a manual or automated move is made Then the move should NOT be added to the record And the record should remain exactly the same	Complete
	11.4	**AC 11.4 <Do not record invalid manual moves>** Given the game is not in replay mode When I attempt to make a manual move And the move is not valid Then the move should NOT be added to the record And the record should remain exactly the same	Complete
	11.5	**AC 11.5 <Do not record automated moves when game is over>** Given the game is not in replay mode And the game is over When I click the automated move button Then the record should remain exactly the same	Complete
	11.6	**AC 11.6 <Wipe the record>** At any time When I press the reset button, change board size, change board type, or randomize Then the record should be wiped clean	Complete
	11.7	**AC 11.7 <Record when randomized>** At any time When I press the randomize button Then the record should be wiped clean And the randomized board should be saved to the record	Complete
12 - Replay Moves	12.1	**AC 12.1 <Replay game>** Given there are moves on the record And I have not randomized since the previous board reset When I press the replay button Then the board should reset And all moves up until now should play on their own, one by one And selected squares should be unselected	Complete

	12.2	**AC 12.2 <Replay from randomized board>** Given there are moves on the record And the board has not been reset since its last randomization When I press the replay button Then the board should revert to its randomized configuration And all the moves until now since then should play on their own, one by one.	Complete
	12.3	**AC 12.3 <Interrupt replay>** Given the game is in replay mode When I change board size, change board type, reset, or randomize Then replay mode should end immediately And no more recorded moves should play And the record should be wiped	Complete
	12.4	**AC 12.4 <Loop replay if already replaying>** Given the game is in replay mode When I press the replay button Then the board should reset And all moves up until now should play on their own, one by one And the previous replay operation should terminate	Complete
	12.5	**AC 12.5 <Don't replay if no moves have been made>** Given that no moves have been recorded When I press the replay button Then nothing should happen And the player should get an alert	Complete
	12.6	**AC 12.6 <Resume game when replay ends>** Given that there are moves on the record And we are in replay mode When all moves on the record are made in replay mode And the moves have not ended the game Then the game should return to play mode And new moves should continue to be added to the record	Complete
	12.7	**AC 12.7 <No alerts if replay ends game>** Given that we are in replay mode When all moves on the record are made in replay mode And there are no more valid moves Then the game should NOT show the usual game over alert And the game should return to play mode	Complete
	12.	**AC 12.8 <Don't allow cell selection in replay mode>**	Co

	8	Given the board is in replay mode When the player attempts to click a cell Then no cell should be selected And the board state should remain the same And replay should continue uninterrupted	mpl ete
	12. 9	**AC 12.9 <Don't allow autoplay in replay mode>** Given the board is in replay mode When the player clicks the autoplay button Then nothing should happen And replay should continue uninterrupted	Co mpl ete

Summary of Source Code

Source code file name	**Production code or test code?**	**# lines of code**
AutomatedGame.java	Production code	48
GameSession.java	Production code	105
ManualGame.java	Production code	79
Board.java	Production code	398
CellState.java	Production code	6
DiamondBoard.java	Production Code	34
EnglishBoard.java	Production Code	42
GameMode.java	Production Code	18
GameScribe.java	Production Code	170
HexagonBoard.java	Production Code	83
Move.java	Production Code	44
Square.java	Production Code	45
BoardView.java	Production Code	288
CellStateTest.java	Test Code	10
DiamondBoardTest.java	Test Code	39

EnglishBoardTest.java	Test Code	354
GameScribeTest.java	Test Code	285
HexagonBoardTest.java	Test Code	73
SquareTest.java	Test Code	42
	Total	2,163

Production Code vs. User Acceptance Criteria

User Stories

User Story ID	**User Story**
1	Choose board type and size
2	Beginning a game
3	View the current board state
4	Make a move
5	Detect game over
6	Determine performance
7	Reset
8	Reconfigure board
9	Automated moves
10	Randomize board
11	Save moves
12	Replay moves

User Story ID and Name	**AC ID**	**Class Name(s)**	**Method Name(s)**	**Status (complete or not)**	**Notes**
1: Choose	1.1	BoardView	listener for	Complete	

board type and size			radio button group within start() method		
	1.2	BoardView	setOnAction for comboBox within start() method	Compete	
	1.3	BoardView	-setOnAction for comboBox within start() method	Complete	ComboBox limits size choices to valid ones
	1.4	BoardView	BoardView class definition, drawBoard	Complete	When we first created BoardView, we create a new board with default configuration s and display it.
2: Beginning a game	2.1	See 1.4	See 1.4	Complete	See 1.4
	2.2	--	--	Cancelled	No longer necessary, as there is no main menu
3: View the current board state	3.1	BoardView	canvas.setOnMouseClicked, autoplayButton.setOnAction, drawBoard	Complete	Mouse clicks and the autoplay button both redraw the board after a move is made
4: Make a move	4.1	BoardView, ManualGame, AutomatedGame,	canvas.setOnMouseClicked, autoplayButton.setOnActi	Complete	

		Board	on, handleClick, makeRandomMove, makeMove		
	4.2	See 4.1	See 4.1	Complete	
	4.3			Complete	
	4.4			Complete	
	4.5			Complete	
	4.6			Complete	
	4.7			Complete	
	4.8			Complete	
	4.9			Complete	
5: Detect game over	5.1	Board	validMovesExist	Complete	
	5.2	--	--	Cancelled	Note: There is no reason for the game to ever initialize without a valid move being possible
6: Determine performance				Not complete	
7: Reset	7.1	BoardView, GameSession, Board	resetButton.setOnAction, handleResetButton resetBoard	Complete	
	7.2	7.1	See 7.1		
8: Reconfigure	See user story 1	See user story 1	See user story 1	Complete	See user story 1

board					
9: Automated moves	9.1	BoardView, automatedController, Board	autoplayButton.setOnAction, makeRandomMove, getRandomMoves, validMovesExist	Complete	
	9.2	BoardView, automatedController, Board	makeRandomMove, getRandomMoves, validMovesExist	Complete	
10: Randomize board	10.1	BoardView, Board	randomButton.setOnAction, randomize	Complete	
	10.2	BoardView, Board	randomButton.setOnAction, randomize	Complete	
	10.3	BoardView, Board	randomButton.setOnAction, randomize	Complete	
11: Save Moves	11.1	GameScribe, ManualGame, GameSession	writeMove, handleClick, GameSession constructor	Complete	
	11.2	GameScribe, AutomatedGame, GameSession	writeMove, makeRandomMove, GameSession constructor	Complete	
	11.3	GameScribe	writeMove	Complete	
	11.4	--	--	Complete	See 11.1
	11.5	--	--	toDo	See 11.2

	11.6	GameSession, BoardView	handleResetButton, handleRandomize, switchBoard	Complete	
	11.7	GameSession, GameScribe, Board	handleRandomize, writeStringToFile, stateToString	Complete	
12: Replay moves	12.1	BoardView, Board, GameScribe	replayButton.setOnAction, replayTimeLine keyframe, makeMoveFromFile, listOfLines	Complete	
	12.2	BoardView, Board, GameScribe	replayButton.setOnAction, loadFromFile, isRandomOnFile	Complete	
	12.3	BoardView	replayTimeLine keyframe, replayButton.setOneAction	Complete	
	12.4	See 12.1	See 12.1	Complete	
	12.5	BoardView, GameScribe	replayButton.setOnAction, haveMovesBeenMade	Complete	
	12.6	See 12.1	See 12.1	Complete	
	12.7	See 12.1	See 12.1	Complete	
	12.8	ManualGame	handleClick	Complete	
	12.9	AutomatedGame	makeRandomMove	Complete	

Tests vs. User Stories/Acceptance Criteria

Automated tests directly corresponding to the acceptance criteria of the above user stories

User Story ID and Name	Acceptance Criterion ID	Class Name (s) of the Test Code	Method Name(s) of the Test Code	Status (Pass/Fail/In Progress)
1 -- Choose board type and size				
2 -- Beginning a game	2.1	EnglishBoardTest	testBoardSize testSizeSevenBoard	Pass
...	2.2	HexagonBoardTest, DiamondBoardTest	testHexBoardSize testSizeSevenHexBoard testSizeFifteenHexBoard testDiBoardSize testSizeSevenDiBoard testSizeFifteenDiBoard	Pass
3 -- View the current board state	3.1	SquareTest, CellStateTest	testInitializePeg testInitializeEmpty testInitializeInvalid testSetState testRow testColumn testEqualsThreeStates	Pass
4 -- Make a move	4.1	EnglishBoardTest	testValidOrthogonal testValidDiagonal testValidDiagonal2 testJumpRemov	Pass

			esPeg testDiagJumpRemovesPeg	
...	4.2	EnglishBoardTest	testInvalidStart testEmptyStart	Pass
...	4.3	EnglishBoardTest	testEmptyJump testEmptyDiagJump	Pass
...	4.4	EnglishBoardTest	testOccupiedLanding	Pass
...	4.5	EnglishBoardTest	testDoubleHop testLongMove	Pass
...	4.6	EnglishBoardTest	testInvalidLanding testOutOfBounds	Pass
...	4.9	EnglishBoardTest	testNeighborMove	Pass
5 -- Detect game over	5.1	EnglishBoardTest	testIfNoValidMoves testDetectValidMoves	Pass
6 -- Determine performance				
7 -- Reset	7.1, 7,2	EnglishBoardTest	resetBoard	Pass
8 -- Reconfigure board				
9 -- Automated Moves				
10 -- Randomize Board	10.2	EnglishBoardTest	testDontRandomizeInvalid	Pass
11 -- Save moves	11.1	GameScribeTest	testTranscribeManual	Pass

	11.2	GameScribeTest	testTranscribeAutomated	Pass
	11.3	GameScribeTest	testDontWriteDuringReplay	Pass
	11.4	GameScribeTest	dontTranscribeInvalidMoves	Pass
	11.5	GameScribeTest	dontRecordAutomatedIfGameOver	Pass
	11.6	GameScribeTest	testRecordWipe	Pass
	11.7	GameScribeTest	testRecordRandomizedState	Pass
12 -- Replay moves	12.1	EnglishBoardTest	testMakeMoveFromFile	Pass
	12.2	EnglishBoardTest, GameScribeTest	testLoadBoardFromFile, testIsRandomBoardOnFile	Pass
	12.5	GameScribeTest	testHaveMovesBeenMade	Pass

Manual tests directly corresponding to some acceptance criteria

User Story ID	Acceptance Criterion ID	Test Case Input	Test Oracle (Expected Output)	Notes
1	1.1-1.3	See test inputs for user story 8	See test oracles for user story 8	I ultimately nixed the “main menu” concept because it was simpler and more usable to simply have a combo box for size and a set of radio buttons for type which update the board

				in real time as they are modified.
	1.4	Starting a new game	Board initializes as a size 7 English board	Success
8	8.1	Changing size or board type during an active game	Board resets itself, updating its configuration.	Success
	8.2	Changing size or board type when a game is over	Board resets itself, updating its configuration	
9	9.1	Press autoplay button when game is not over	A random valid move is made	Repeated trials showed random results. There were no occasions where it failed to make a move (unless the game was over) and no moves made that were invalid.
	9.2	Press autoplay button when game is over	No move is made	Success
10	10.1	Pressed randomize button	All cells are randomly empty or pegged	Repeated trials showed random results. Resetting the game did *not* cause the same configurations to appear, so it was truly random as desired.
	10.3	Modified code so only one in fifty squares had	A randomized board where at least one valid	The way I guaranteed the board will have

		a peg in them, then pressed randomized button	move is possible.	valid move(s) was by having the randomize function loop until it creates a board with one or more valid moves available. When only one square in fifty has a peg, the chance of invalid boards being created is much higher, so the 100% valid boards I saw confirm that 10.3 is implemented.
11 -- Save moves	11.2	Clicked autoplay button a few times and observed which moves were printed out in the system, then compared them to what was on file.	System print outs that matched the moves I was observing, and a file that matched the system print outs.	The automated tests for AC 11.2 only counts the number of entries recorded to the file when three automated moves are made; this test was necessary to make sure those entries were accurate.
	11.6	Made a few moves and changed type/size and checked if the record was wiped	Empty file after switch	Record automatically wiped since BoardView is creating a new GameSession object, and therefore new Scribe object with new file.

12 -- Replay moves	12.1	Made a few moves and pressed replay. Then, pressed replay while a replay was already in progress. Then, made some moves, selected a peg, pressed replay, let the replay finish, and clicked on an empty hole that would have been a valid move from the peg I had selected.	Board resets and replays all the moves I made, one-by-one. Replay starts over and continues. Peg I selected is de-selected on the GUI, and the move does *not* succeed because the square is no longer selected.	I was able to automate tests to make sure it was possible to make moves from a file, but I had to do manual tests to make sure the threads and real-time play were working properly.
	12.2	Randomized board, made a few moves, and pressed the replay button. Follow up manual tests similar to 12.1	Board resets to randomized configuration and the moves played after that are replayed one by one. Follow up manual results similar to 12.1	Automated tests validated loadFromFile function as well as the function to detect if a random board existed on the file; manual tests needed to see the feature work in real time.
	12.3	Made a few moves, started replay mode, and then interrupted replay by switching size, switching type, resetting, and pressing	In all cases, thread should immediately end (signified by "We have reached the end!" message in console).	No automatic test since this AC is all about thread behavior.

		randomize.		
	12.4	Press repay button while autoplay is in progress, multiple times.	Thread should start over and continue without delay each time.	No automatic test since this AC is all about thread behavior.
	12.5	Press replay during new game, right after a reset, or right after a randomize	The thread should not run at all, and an alert should pop up.	
	12.6	Made some moves, replayed, made some more, replayed, made some automatic moves, replayed.	Should be able to make moves as soon as each replay has finished, and each replay should include the new moves made since the last one.	
	12.7	Played game to completion, got alert, then replayed the whole thing.	No alert should play when replay finishes.	
	12.8	Entered replay mode and then clicked wherever I could. Also made sure to attempt valid moves during replay.	No cell selection occurs, and no moves get made.	
	12.9	Entered replay mode and clicked autoplay mode wildly.	No random moves should be made until replay ends, and the game state should not be	

			effected at all by autoplay button clicks.	